

Integration WS 2

Evaluate Q1-10

1. $\int \sin^2 x \cos^3 x \, dx$

2. $\int \cos^2 x \sin^5 x \, dx$

3. $\int \frac{\cos^3 x}{\sin^2 x} \, dx$

4. $\int \frac{\sin^3 x}{\cos^5 x} \, dx$

5. $\int \sqrt[3]{\sin x} \cos^3 x \, dx$

6. $\int \cos 6x \cos 2x \, dx$

7. $\int \sin 6x \sin 2x \, dx$

8. $\int \sin 3x \cos x \, dx$

9. $\int \cos 3x \sin x \, dx$

10. $\int \cos 4x \cos 2x \, dx$

11. a. Why is it not valid to evaluate $\int_0^2 \sqrt{\frac{x}{4-x}} \, dx$

using the techniques of this section?

b. Nevertheless, show that its value is

$$\lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^2 \sqrt{\frac{x}{4-x}} \, dx = \pi - 2$$

12. a. Show that $x^3 + 3x^2 + 5x + 1 = (x+1)(x^2 + 2x + 2) + (x-1)$.

b. Hence or otherwise show that

$$\int_{-1}^0 \frac{x^3 + 3x^2 + 5x + 1}{\sqrt{x^2 + 2x + 2}} \, dx = \frac{1}{3}(5\sqrt{2} - 4) - 2 \ln(1 + \sqrt{2})$$

13. Find:

a. $\int x^2 e^x \, dx$

b. $\int x^2 \cos x \, dx$

c. $\int (\ln x)^2 \, dx$

14. Determine:

a. $\int e^x \cos x \, dx$

b. $\int e^{-x} \sin x \, dx$

15. Evaluate:

a. $\int_0^{\frac{\pi}{2}} e^{2x} \cos x \, dx$

b. $\int_0^{\frac{\pi}{4}} e^x \sin 2x \, dx$.

16. Use integration by parts to evaluate:

a. $\int_0^{\frac{\sqrt{3}}{2}} \sin^{-1} x \, dx$

b. $\int_{\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \cos^{-1} x \, dx$

c. $\int_0^1 4x \tan^{-1} x \, dx$

17. Determine formulae for the following:

a. $\int \sqrt{a^2 - x^2} \, dx$

b. $\int \ln(x + \sqrt{x^2 + a^2}) \, dx$

c. $\int \ln(x + \sqrt{x^2 - a^2}) \, dx$

18. Find:

a. $\int \sin 3x \cos x \, dx$

b. $\int \cos 3x \sin x \, dx$

c. $\int \cos 6x \cos 2x \, dx$

19. Determine:

a. $\int_0^{\frac{1}{2}} x \sin^{-1} x \, dx$

b. $\int_0^1 x^2 \tan^{-1} x \, dx$

20. Given $I_n = \int (1 - \sqrt{x})^n \, dx$

a. Show that $I_n = \frac{2}{n+2} [x(1 - \sqrt{x})^n] + \frac{n}{n+2} I_{n-1}$

b. Hence evaluate $\int_0^1 (1 - \sqrt{x})^6 \, dx$

21. a. Show that $(1-x)^{n-1} - (1-x)(1-x)^{n-1} =$

$$x(1-x)^{n-1}$$

b. If $I_n = \int_0^1 \sqrt{x}(1-x)^n \, dx$ where n is a non-negative integer, show that $I_n =$

$$\frac{2n}{2n+3} I_{n-1}$$

c. Evaluate $\int_0^1 \sqrt{x}(1-x)^3 \, dx$

22. If $I_n = \int_0^2 (x^3 - 8)^n \, dx$ where n is a positive

integer, show that $I_n = \frac{-24n}{3n+1} I_{n-1}$

23. Use integration by parts to evaluate

$$\int_0^{\infty} 3ye^{-y}(1 - e^{-y})^2 \, dy$$

24. a. Explain why the domain of the function,

$$f(x) = \sqrt{2 - \sqrt{x}} \text{ is } 0 \leq x \leq 4$$

b. Show that $f(x)$ is a decreasing function and

hence find its range

c. Using the substitution, $u = 2 - \sqrt{x}$ or

otherwise, find the area bounded by the curve and the x and y axes

25. a. Show that $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{x(\pi-2x)} = \frac{2}{\pi} \ln 2$

b. Using the substitution $u = a + b - x$, show

$$\text{that } \int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

c. Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^2 x}{x(\pi-2x)} dx$

26. Let $I_{2n} = \int_{-1}^1 (1 - x^2)^n dx$ where $n \geq 0$

a. Use the substitution $x = \sin \theta$ to show

$$\text{that } I_{2n} = \frac{2n}{2n+1} I_{2n-2}$$

b. Show that $I_6 = \frac{32}{35}$

c. Deduce that $I_{2n} = \frac{2^{2n+1}(n!)^2}{(2n+1)!}$

27. a. Show that $f(x) = \frac{\sec x + \tan x}{2 \sec x + 3 \tan x}$ is a decreasing

function in terms of x for the domain

$$0 < x < \frac{\pi}{2}$$

b. Deduce that $\frac{\pi}{28} > \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sec x + \tan x}{2 \sec x + 3 \tan x} dx >$

$$\frac{(\sqrt{2}-1)\pi}{12}$$

28. a. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$ then show that

$$I_n = \frac{n-1}{n} I_{n-2}$$

b. By first finding I_5 , I_7 and I_9 in unsimplified form, and noting a pattern in your answers,

$$\text{show that } I_n = \frac{\left\{\left[\frac{1}{2}(n-1)\right]!\right\}^2}{n!} 2^{n-1} \text{ if } n \text{ is odd}$$

29. a. By rewriting $\cos(n+2)x$ as $\cos\{(n+1)+1\}x$ and $\cos nx$ as $\cos\{(n+1)-1\}x$, show that $\cos(n+2)x + \cos nx = 2 \cos(n+1)x \cos x$

b. Hence prove that given $u_n = \int_0^{\frac{\pi}{2}} \frac{1 - \cos nx}{1 - \cos x} dx$

where n is a positive integer or zero, then,

$$u_{n+2} + u_n - 2u_{n+1} =$$

$$\int_0^{\frac{\pi}{2}} \frac{2 \cos(n+1)x \{1 - \cos x\}}{1 - \cos x} dx = 0$$

c. Evaluate u_0 and u_1 directly, and hence evaluate u_2 and u_3

d. Also show that $\int_0^{\frac{\pi}{2}} \frac{\sin^2 3\theta}{\sin^2 \theta} d\theta = \frac{3\pi}{2}$

30. Find $\int_{-1}^1 x^2 \sin^7 x dx$, giving reasons

Answers

- $\frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x$
- $-\frac{1}{7} \cos^7 x + \frac{2}{5} \cos^5 x - \frac{1}{3} \cos^3 x$
- $-\sin x - \operatorname{cosec} x$
- $-\frac{1}{2} \sec^2 x + \frac{1}{4} \sec^4 x$
- $\frac{3}{4} (\sin x)^{\frac{4}{3}} - \frac{3}{10} (\sin x)^{\frac{10}{3}}$
- $\frac{1}{16} \sin 8x + \frac{1}{8} \sin 4x$
- $\frac{1}{8} \sin 4x - \frac{1}{16} \sin 8x$
- $-\frac{1}{8} \cos 4x - \frac{1}{4} \cos 2x$
- $-\frac{1}{8} \cos 4x + \frac{1}{4} \cos 2x$
- $\frac{1}{12} \sin 6x + \frac{1}{4} \sin 2x$
- $\frac{x}{\sqrt{4x-x^2}}$ is undefined at $x = 0$.
- a. $(2 - 2x + x^2)e^x + C$; b. $x^2 \sin x + 2x \cos x - 2 \sin x + C$; c. $x(\ln x)^2 - 2x \ln x + 2x + C$
- a. $\frac{1}{2} e^x (\cos x + \sin x) + C$; b. $-\frac{1}{2} e^{-x} (\cos x + \sin x) + C$
- a. $\frac{1}{5} (e^{\pi} - 2)$; b. $\frac{1}{5} (e^{\frac{\pi}{4}} + 2)$
- a. $\frac{1}{2\sqrt{3}} (\pi - \sqrt{3})$; b. $\frac{\sqrt{3}\pi}{2}$; c. $\pi - 2$
- a. $\frac{1}{2} (x\sqrt{a^2 - x^2} + a^2 \sin^{-1}(\frac{x}{a})) + C$; b. $x \ln(x + \sqrt{x^2 + a^2}) - \sqrt{x^2 + a^2} + C$; c. $x \ln(x + \sqrt{x^2 - a^2}) - \sqrt{x^2 - a^2} + C$
- a. $-\frac{1}{8} \cos 4x - \frac{1}{4} \cos 2x + C$; b. $-\frac{1}{8} \cos 4x + \frac{1}{4} \cos 2x + C$; c. $\frac{1}{16} \sin 8x + \frac{1}{8} \sin 4x + C$
- a. $\frac{1}{48} (3\sqrt{3} - \pi)$; b. $\frac{1}{12} (\pi + 2 \ln 2 - 2)$
- b. $\frac{1}{28}$
- c. $\frac{32}{315}$
- $\frac{11}{6}$
- b. Range: $0 \leq y \leq \sqrt{2}$; c. $A = \frac{32}{15} \sqrt{2} \text{ units}^2$
- c. $\frac{1}{\pi} \ln 2$
- c. $u_0 = 0, u_1 = \pi, u_2 = 2\pi, u_3 = 3\pi$
- 0